DEPARTMENT OF TECHNICAL EDUCATION

GOVERNMENT OF KARNATAKA



450 - R.E.S. POLYTECHNIC, BANGALORE

PROGRAMMING CONCEPTS IN MECHANICAL ENGINEERING ACTIVITIES

Course Code: 25ME34I | Semester III | Mechanical Engineering

Submitted By:

Name: _____

Register Number: _____

RUBRICS FOR ASSESSMENT OF ACTIVITIES AND RECORD BOOK

Sl. No.	Dimension	Unsatisfactory (1-10)	Need Improvement (11-20)	Satisfactory (21-30)	Good (31-40)	Excellent (41-50)	Student Score
01	Understanding Activity Question	Do not understand at all	Poor understanding	Little understanding	Good understanding	Excellent understanding	
02	Execution of Activities	No	Major errors	Inaccuracy	Effective but minor errors	Precise and accurate	
03	Record Book Maintenance	No quality	Major flaws	Acceptable	Good quality	High quality	
04	In time submission of record	Major sections not addressed / poor	Takes time, 20% submitted in time	50% submitted in time	80% submitted in time	100% submitted in time	
Average Marks							

PROGRAMMING CONCEPTS IN MECHANICAL ENGINEERING

One Activity per Week (Week 1 to Week 13) - Aim, Algorithm and Result

Name: _____ Reg. No.: _____ Section: _____

WEEK 1: Algorithms and Flow Charts

Activity 1: Finding the maximum of three forces (Decision)

Aim:

To write the algorithm and flowchart to find the maximum of three forces.

Algorithm:

1. Start.
2. Read forces F1, F2 and F3.
3. If F1 >= F2 and F1 >= F3, then Max = F1.
4. Else if F2 >= F3, then Max = F2, else Max = F3.
5. Print Max.
6. Stop.

Flowchart:

Result:

Algorithm and flowchart written. Example: 40 N, 65 N, 52 N gives Max = 65 N.

WEEK 2: Introduction to Python and Basic Constructs

Activity 2: Output speed and torque of a gear system

Aim:

To calculate output speed and torque of a gear system for user-inputted input speed, input torque and gear ratio.

Algorithm:

1. Start.
2. Read input speed N1, input torque T1 and gear ratio G.
3. Calculate output speed N2 = N1 / G.
4. Calculate output torque T2 = T1 x G.
5. Print N2 and T2.
6. Stop.

Result:

Program executed. Example: N1 = 1500 rpm, T1 = 10 N-m, G = 3 gives N2 = 500 rpm and T2 = 30 N-m (ideal case).

WEEK 3: Control Structures (Conditional Statements and Loops)

Activity 3: Factorial of a number using while loop

Aim:

To write a Python program to calculate the factorial of a number using a while loop.

Algorithm:

1. Start.
2. Read n. Set fact = 1.
3. While n > 1: fact = fact x n and n = n - 1.
4. Print fact.
5. Stop.

Result:

Program executed. Example: 5! = 120.

WEEK 4: Functions and Modules

Activity 4: GCD of two numbers using a function

Aim:

To find the greatest common divisor (GCD) of two numbers using a function.

Algorithm:

1. Start.
2. Define gcd(a, b).
3. While b != 0: set a, b = b, a % b.
4. Return a.
5. Read two numbers, call gcd() and print the result.
6. Stop.

Result:

Function executed. Example: GCD(12, 18) = 6.

WEEK 5: Data Structures (Lists, Tuples and Dictionaries)

Activity 5: Dictionary for student records (name, age, grade)

Aim:

To create a dictionary of student records and write functions to add and remove entries.

Algorithm:

1. Start.
2. Create a dictionary with name, age and grade as keys.
3. Define add_entry() to add a key-value pair.
4. Define remove_entry() to delete a key using del or pop().
5. Call the functions and print the dictionary.
6. Stop.

Result:

Program executed. Records were added and removed successfully.

WEEK 6: File Handling and Exception Handling

Activity 6: Read a CSV file and count the number of rows

Aim:

To read data from a CSV file and count the number of rows.

Algorithm:

1. Start.
2. Import the csv module and open the CSV file.
3. Read the file using csv.reader().
4. Count the rows using a loop or len(list(reader)).
5. Print the number of rows.
6. Stop.

Result:

Program executed. The number of rows in the CSV file was displayed.

WEEK 7: Object-Oriented Programming (OOP)

Activity 7: Bank account class

Aim:

To develop a Python class for a bank account with deposit, withdrawal and balance inquiry.

Algorithm:

1. Start.
2. Create class Account with balance in __init__.
3. Define deposit(amount) to add to balance.
4. Define withdraw(amount); allow only if balance is enough.
5. Define balance inquiry method.
6. Create object, perform operations and print balance.
7. Stop.

Result:

Program executed. Example: balance 1000, deposit 500, withdraw 300 gives balance 1200.

WEEK 8: Raspberry Pi: Motor Control and Automation

Activity 8: Controlling a DC motor using L298N motor driver with Raspberry Pi

Aim:

To control a DC motor using the L298N motor driver and Raspberry Pi.

Algorithm:

1. Connect motor to L298N output; connect IN1, IN2 and ENA to GPIO pins.
2. Import RPi.GPIO and set the pins as output.
3. IN1 = HIGH, IN2 = LOW: motor runs forward. Reverse the levels for backward.
4. Use PWM on ENA to change speed.
5. Set all pins LOW to stop and call GPIO.cleanup().

Result:

The DC motor ran forward and backward with speed control.

WEEK 9: Raspberry Pi: Sensor Integration and Data Collection

Activity 9: Building a simple robot using Raspberry Pi, motors and sensors

Aim:

To build a simple robot using Raspberry Pi, motors and sensors.

Algorithm:

1. Connect two DC motors through the L298N driver and connect a sensor to the Raspberry Pi.
2. Read the sensor value continuously.
3. If no obstacle, move the robot forward.
4. If an obstacle is detected, stop and turn.
5. Repeat the loop.

Result:

The simple robot moved and changed direction according to the sensor input.

WEEK 10: Statistical Quality Control: X-bar and R Charts

Activity 10: Sample average, sample range and X-bar and R charts using Matplotlib

Aim:

To calculate Sample Average (X-bar) and Sample Range (R) for given data and plot X-bar and R charts using Matplotlib.

Algorithm:

1. Start. Enter the sample data (each row is one sample).
2. Calculate X-bar = mean and R = max - min for each sample.
3. Calculate X-double-bar (average of X-bars) and R-bar (average of ranges).
4. X-bar chart: UCL = X-double-bar + A2 x R-bar, LCL = X-double-bar - A2 x R-bar.
5. R chart: UCL = D4 x R-bar, LCL = D3 x R-bar.
6. Plot both charts using Matplotlib with centre line and limits.
7. Stop.

Result:

X-bar and R charts were plotted successfully using Matplotlib.

WEEK 11: Introduction to Statistical Quality Control (SQC)

Activity 11: Identify in-control and out-of-control points

Aim:

To identify in-control and out-of-control points on the control charts.

Algorithm:

1. Start.
2. Calculate UCL and LCL.
3. For each sample point, check if LCL <= point <= UCL.
4. If yes, mark as in control; else mark as out of control (plot in red).
5. Print the out-of-control sample numbers.
6. Stop.

Result:

Out-of-control points were identified. If all points lie within the limits, the process is in control.

WEEK 12: Hypothesis Testing and TQM

Activity 12: Plot different quality control tools for the given data

Aim:

To generate Python code to plot quality control tools (Pareto chart, fishbone diagram and flowchart) for the given data.

Algorithm:

1. Start. Enter defect types and their counts.
2. Sort the counts in descending order and find the cumulative percentage.
3. Plot a bar chart with the cumulative percentage line (Pareto chart).
4. Draw the fishbone (cause and effect) diagram using lines and text in Matplotlib.
5. Draw the process flowchart using boxes and arrows.
6. Show the plots. Stop.

Result:

The quality control tools were plotted. The Pareto chart shows the few causes that create most of the defects.

WEEK 13: Finite Element Methods (FEM)

Activity 13: Compare stress distribution in the plate with and without hole

Aim:

To compare and plot the stress distribution in the plate with and without a hole.

Algorithm:

1. Start. Read load P, width w, thickness t and hole diameter d.
2. Without hole: stress = P / (w x t).
3. With hole: net area = (w - d) x t; nominal stress = P / net area; maximum stress = Kt x nominal stress.
4. Plot both stress distributions on the same graph.
5. Compare. Stop.

Result:

The plate with a hole shows higher stress near the hole (stress concentration) than the plate without a hole.

Student Signature	Staff Signature	HOD Signature